

P2: Two-Loop Running, Spectrum Thresholds, Covariance, and the PQ Cosmology Repair

Route-F four-dimensional mainline

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Abstract

We execute the P2 gate on the complete 35-row scalar spectrum of P54PQ-v2. The two-loop $G_{\text{SM}} \rightarrow SU(4)_c \times SU(2)_L \times SU(2)_R \rightarrow \text{Spin}(10)$ engine reproduces the published no-threshold reference scales, and the one-loop threshold Jacobian uses all 290 positive physical scalar directions. The original $\sigma/\omega = 0.35$ diagnostic fails, but re-deriving only the radial quadratic masses and retaining every cubic/quartic P1 coupling produces a stable one-doublet branch. Exact Pati–Salam Casimir projectors and spectral traces then give the self-consistent threshold fixed point $\sigma/\omega = 0.1265$, $M_I/M_U = 0.126525$. Two-site scale covariance and the loop-level one-doublet renormalization condition are closed. The follow-up Landau- $\overline{\text{MS}}$ hard-mode calculation gives the bosonic matching curvature $\Pi_B/\omega^2 = -0.0202366510950\,1_4$; the heavy-Yukawa projection is retained as an explicit nuisance rather than set to zero. In parallel, two Weyl 10_F fields of PQ charge +2 give a minimal candidate extension with physical $N_{\text{DW}} = 1$.

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1 Scope, conventions, and promotion boundary

The input action is the corrected P54PQ-v2 action. The historical P1 stationary point

$$(\omega, \sigma, v_s) = (1, 0.35, 0.25). \tag{1}$$

The scalar Hessian is not replaced by an extended-survival ansatz: every threshold trace below is computed from its 35 Standard-Model Casimir rows. The two-loop coefficients and finite matching

convention follow the general product-group RGE and one-loop decoupling formulas used in Refs. [1, 2, 3]. Low-energy central values are updated to

$$M_Z = 91.1876 \text{ GeV}, \quad \hat{\alpha}^{-1}(M_Z) = 127.955, \quad \hat{s}_W^2 = 0.23122, \quad \alpha_s(M_Z) = 0.1180, \quad (2)$$

with the last value taken from the 2024 PDG review [4].

There are two different meanings of “threshold calculation” that must not be conflated:

1. the lower-group trace $\text{tr } t_i^2 \log(M/\mu)$ of a physical mass eigenstate is defined once its G_{SM} representation and mass are known;
2. assigning that state to the M_I or M_U matching functional requires a parametrically separated EFT interval and its parent G_{PS} multiplet.

The historical point supplies the first datum but not the second. We retain it as a fail-closed regression in Sec. 5; the new fixed point in Sec. 6 supplies both. The bosonic zero-momentum matching coefficient is now closed in a declared scheme. A gauge-independent pole mass remains outside this artifact because it needs momentum dependence and the globally fitted heavy Yukawas.

2 Two-loop running and finite matching

Write $t = \log \mu$ and $\alpha_i = g_i^2/(4\pi)$. In each effective theory,

$$\frac{d\alpha_i^{-1}}{dt} = -\frac{a_i}{2\pi} - \sum_j \frac{b_{ij}}{8\pi^2 \alpha_j^{-1}}. \quad (3)$$

Below M_I the ordering is $(3_c, 2_L, 1_Y)$, with GUT-normalized $g_1^2 = 5g_Y^2/3$, and

$$a^{\text{SM}} = \left(-7, -\frac{19}{6}, \frac{41}{10} \right),$$

$$b^{\text{SM}} = \begin{pmatrix} -26 & 9/2 & 11/10 \\ 12 & 35/6 & 9/10 \\ 44/5 & 27/10 & 199/50 \end{pmatrix}. \quad (4)$$

Above M_I the ordering is $(4_c, 2_L, 2_R)$ and the D-parity spectrum gives

$$a^{\text{PS}} = \left(1, \frac{26}{3}, \frac{26}{3} \right),$$

$$b^{\text{PS}} = \begin{pmatrix} 1209/2 & 249/2 & 249/2 \\ 1245/2 & 779/3 & 48 \\ 1245/2 & 48 & 779/3 \end{pmatrix}. \quad (5)$$

These matrices include the intermediate scalar content specified by the traditional extended-survival Pati–Salam baseline. Their role here is to determine the interval demanded by low-energy data; they are not silently identified with the nonhierarchical P1 spectrum.

For a simple group G broken to G_i , one-loop $\overline{\text{MS}}$ matching is [2]

$$\alpha_i^{-1}(\mu) = \alpha_G^{-1}(\mu) - \frac{\lambda_i(\mu)}{12\pi}, \quad (6)$$

where

$$\lambda_i = (C_G - C_i) - 21 \text{tr} \left[t_{iV}^2 \log \frac{M_V}{\mu} \right] + \text{tr} \left[t_{iS}^2 \mathcal{P}_{\text{PGB}} \log \frac{M_S}{\mu} \right] + 8 \text{tr} \left[t_{iF}^2 \log \frac{M_F}{\mu} \right]. \quad (7)$$

The projector removes eaten Goldstones. With logarithmic thresholds set to zero, the finite Casimir terms give

$$\begin{aligned}\alpha_4^{-1}(M_I^+) &= \alpha_3^{-1}(M_I^-) + \frac{1}{12\pi}, \\ \alpha_{2L}^{-1}(M_I^+) &= \alpha_2^{-1}(M_I^-), \\ \alpha_{2R}^{-1}(M_I^+) &= \frac{5}{3}\alpha_1^{-1}(M_I^-) - \frac{2}{3}\alpha_4^{-1}(M_I^+) + \frac{14}{36\pi}.\end{aligned}\tag{8}$$

At M_U , the three implied unified inverse couplings are

$$\alpha_{10,a}^{-1} = \alpha_a^{-1} + \frac{8 - C_a}{12\pi}, \quad (C_4, C_{2L}, C_{2R}) = (4, 2, 2).\tag{9}$$

Equating them supplies two nonlinear equations for $(\log M_I, \log M_U)$.

Using the historical inputs of Ref. [2], the implementation returns

$$M_I = 4.61890 \times 10^{13} \text{ GeV}, \quad M_U = 1.21714 \times 10^{15} \text{ GeV},\tag{10}$$

reproducing the published 4.62×10^{13} and 1.22×10^{15} GeV values. With Eq. (2),

$M_I = 4.60943 \times 10^{13} \text{ GeV}, \quad M_U = 1.18126 \times 10^{15} \text{ GeV}, \quad \alpha_U^{-1} = 37.60198.$

(11)

Thus

$$\left(\frac{M_I}{M_U}\right)_{2L} = 0.0390213, \quad \log \frac{M_U}{M_I} = 3.24365.\tag{12}$$

3 The exact 35-sector scalar threshold trace

For row k , let d_k be its real multiplicity, $C_{3,k}$ and $C_{2,k}$ its Casimirs, and Y_k^2 its hypercharge square. The trace on canonical real coordinates is

$$\mathbf{c}_k = \left(\frac{d_k C_{3,k}}{8}, \frac{d_k C_{2,k}}{3}, \frac{3}{5} d_k Y_k^2 \right).\tag{13}$$

This automatically gives $2T(R)$ for a complex representation written as $R \oplus \bar{R}$ and $T(R)$ for a real scalar. Therefore no ad hoc “complex-factor” is attached to individual rows.

As a normalization theorem, summing all 328 coordinates gives

$$\sum_{k=1}^{35} \mathbf{c}_k = (84, 84, 84).\tag{14}$$

This is exactly the index of $54_{\mathbb{R}} \oplus 126_{\mathbb{C}} \oplus 10_{\mathbb{C}}$ restricted to each canonically normalized Standard-Model factor. Removing 33 gauge Goldstones, the PQ Goldstone, and the four real light-doublet coordinates leaves 290 positive directions in 29 rows, with

$$\sum_{k \in +} \mathbf{c}_k = (79, 77, 75.4).\tag{15}$$

Their masses occupy

$$0.116278 \omega \leq M_{S,k} \leq 1.682012 \omega.\tag{16}$$

At the single reference $\mu = \omega$, the fully correlated scalar trace is

$$\lambda_S(\omega) = \sum_{k \in +} \mathbf{c}_k \log \frac{M_{S,k}}{\omega} = (-56.89360, -56.33640, -53.16705).\tag{17}$$

Equation (17) is a valid lower-group trace. It is not yet a prescription for distributing rows between Eqs. (8) and (9).

4 Covariance and what it does not prove

Let $\mathbf{y} = (\log_{10} M_I, \log_{10} M_U, \alpha_U^{-1})$ and let the two unification equations be $\mathbf{F}(\mathbf{y}, \mathbf{x}) = 0$. The implicit-function Jacobian is

$$J_{yx} = - \left(\frac{\partial \mathbf{F}}{\partial \mathbf{y}} \right)^{-1} \frac{\partial \mathbf{F}}{\partial \mathbf{x}}, \quad \Sigma_y = J_{yx} \Sigma_x J_{yx}^T. \quad (18)$$

For low-energy errors $(0.010, 3 \times 10^{-5}, 9 \times 10^{-4})$ in $(\alpha_{\text{em}}^{-1}, s_W^2, \alpha_s)$, the propagated standard deviations are

$$\sigma_y^{\text{exp}} = (0.00794, 0.02557, 0.06226). \quad (19)$$

The JSON artifact also exports the exact 3×29 matrix with columns $\mathbf{c}_k$. To test the covariance machinery, not to assert a physical prior, we assign independent 10% log-mass errors and collapse all rows at M_I through

$$\delta \alpha^{-1} = - \frac{1}{12\pi} \sum_{k \in +} \mathbf{c}_k \delta \log M_k. \quad (20)$$

This gives

$$\sigma_y^{\text{thr, coll}} = (0.02102, 0.01492, 0.07812), \quad \sigma_y^{\text{total, coll}} = (0.02247, 0.02961, 0.09989). \quad (21)$$

The threshold covariance is positive semidefinite and retains all off-diagonal correlations. However, “coll” is an essential qualifier: matching-scale covariance cannot cure a missing EFT interval.

5 Historical hierarchy obstruction

The physical vector spectrum in units of $g_{10}^2 \omega^2$ is

$$0 \times 12, \quad 0.1225 \times 8, \quad 0.6125 \times 1, \quad 1 \times 12, \quad 1.1225 \times 12. \quad (22)$$

The first eight massive vectors have $M_I/(g\omega) = 0.35$ while the first unification band has $M_U/(g\omega) = 1$. Hence the action itself fixes

$$\left(\frac{M_I}{M_U} \right)_{\text{P1}} = \frac{\sigma}{\omega} = 0.35, \quad (23)$$

a factor

$$\frac{0.35}{0.0390213} = 8.96946 \quad (24)$$

above the RGE-preferred ratio. If this ratio is frozen and only the overall scale is varied, the least-squares point is

$$M_I = 1.33989 \times 10^{14} \text{ GeV}, \quad M_U = 3.82827 \times 10^{14} \text{ GeV}, \quad (25)$$

but the two independent unified-coupling residuals are

$$(\Delta_{4L}, \Delta_{4R}) = (-2.06910, 0.43300), \quad \|\Delta\|_2 = 2.11392. \quad (26)$$

This is a mismatch in inverse couplings, not a numerical integration error.

At the no-threshold solution $g_U = 0.57810$, the representative vector bands are 0.20233ω and 0.57810ω . Equation (16) crosses both. Thus there is neither a large logarithmic Pati–Salam interval nor a disjoint scalar allocation to two matching surfaces. Applying all 29 rows at whichever surface improves a fit would be a scheme-dependent assignment, not one-loop EFT matching.

The necessary repair is therefore structural:

rebuild a stationary P1 branch near $\sigma/\omega \simeq 0.039$, then recompute its full Hessian.

(27)

Only that new spectrum can decide whether threshold corrections preserve or destroy the hierarchy. The following section performs this calculation; therefore Eq. (27) is now a completed action item, not the current blocker.

6 Parent-resolved threshold fixed point

Set $r = \sigma/\omega$. All cubic and quartic couplings are held at their P1 values. At each r the three radial quadratic coefficients are fixed by the stationarity equations. The fixed point is

$$r = 0.1265, \quad (\mu^2, \nu^2, \mu_s^2) = (1.40952534, 0.38190203, 0.17192027) \omega^2, \quad (28)$$

and the unique first physical crossing occurs at $\xi_0^2 = 0.147371094 \omega^2$. The complete Hessian has 38 zero modes—33 gauge, one PQ, and four real coordinates of a single Higgs doublet—and no negative mode.

Let C_4, C_L, C_R be the three Casimir operators on all 328 real scalar coordinates. Their joint projectors reproduce

$$\begin{aligned} 54 &= (1, 1, 1) \oplus (6, 2, 2) \oplus (20', 1, 1) \oplus (1, 3, 3), \\ 126_{\mathbb{C}} &= (6, 1, 1) \oplus (10, 3, 1) \oplus (15, 2, 2) \oplus (\overline{10}, 1, 3), \\ 10_{\mathbb{C}} &= (6, 1, 1) \oplus (1, 2, 2). \end{aligned} \quad (29)$$

The VEV lies in $(\overline{10}, 1, 3)$ with projector weight $1 - 3.3 \times 10^{-16}$. We therefore define P_I as this exact parent projector and $P_U = 1 - P_I$. No broken-phase state is assigned by its nearest mass. Instead the scalar thresholds are the basis-independent traces

$$\lambda_{I,i}^S = \text{tr}[t_{i,G_{\text{SM}}}^2 P_I \log(M_S/\mu_I)], \quad \lambda_{U,a}^S = \text{tr}[t_{a,G_{\text{PS}}}^2 P_U \log(M_S/\mu_U)]. \quad (30)$$

The 45-dimensional adjoint mass operator gives nine M_I -site and 24 M_U -site massive vectors; their -21 tr terms are evaluated in the same way. With $\mu_I = g_U \sigma$, $\mu_U = g_U \omega$, iteration of g_U and the two matching equations converges to

$$\boxed{M_I = 1.07986 \times 10^{14} \text{ GeV}, \quad M_U = 8.53475 \times 10^{14} \text{ GeV}, \quad \alpha_U^{-1} = 39.70141,} \quad (31)$$

with $M_I/M_U = 0.126524923$. Relative to the input $r = 0.1265$, the replay factor is 1.000197.

The total logarithmic threshold vectors are

$$\lambda_I = (31.21384, 0, 51.86696), \quad \lambda_U = (-8.46162, -12.54752, -3.02191). \quad (32)$$

For independent 10% log-mass errors on each of the 27 exactly degenerate scalar/vector mass blocks, the propagated standard deviations in $(\log_{10} M_I, \log_{10} M_U, \alpha_U^{-1})$ are

$$\begin{aligned} \sigma_{\text{exp}} &= (0.00794, 0.02559, 0.06130), \\ \sigma_{\text{thr}} &= (0.03952, 0.02848, 0.24057), \\ \sigma_{\text{total}} &= (0.04031, 0.03829, 0.24826). \end{aligned} \quad (33)$$

The full covariance matrices and the 6×27 mass-to-matching Jacobian are exported in the machine artifact.

7 Renormalized one-doublet condition

Let $D(\xi_0^2)$ be the tree-level doublet mass matrix and u its normalized light eigenvector. From P1,

$$\|\mathcal{P}_{10}u\|^2 = 0.999997474290, \quad \|\mathcal{P}_{126}u\|^2 = 2.52571049 \times 10^{-6}. \quad (34)$$

Since the action contains $-\xi_0^2 \phi^\dagger \phi$, the Hellmann–Feynman theorem gives

$$\kappa \equiv \frac{\partial m_h^2}{\partial \xi_0^2} = -\langle u, \mathcal{P}_{10}u \rangle = -0.999997474290. \quad (35)$$

For a finite renormalized self-energy $\Pi_D(0; \mu)$, first-order matching of the light eigenvalue gives the unique counter-retuning

$$\delta\xi_0^2 = -\frac{u^\dagger \Pi_D(0; \mu) u}{\kappa} = 1.00000252572 \frac{\Pi_{hh}(0; \mu)}{\omega^2}. \quad (36)$$

The nearest positive doublet eigenvalue is

$$\Delta_D = 0.0305570303 \omega^2. \quad (37)$$

With $Q = 1 - uu^\dagger$, a sufficient single-doublet condition is

$$\left\| Q \left[\Pi_D + \delta\xi_0^2 \frac{\partial D}{\partial \xi_0^2} \right] Q \right\|_2 < \Delta_D. \quad (38)$$

This completes the retuning theorem but not its numerical evaluation. The Coleman–Weinberg Hessian

$$\Delta\mathcal{H}_{IJ} = \partial_I \partial_J \frac{1}{64\pi^2} \text{STr} \left[M^4(\varphi) \left(\log \frac{M^2(\varphi)}{\mu^2} - c_s \right) \right] \quad (39)$$

requires field-dependent scalar, gauge, ghost, and fermion masses. The heavy fermion part also requires the renormalized Y_{10}, Y_{126} matrices that P3 has not yet fit. Substituting only the vacuum eigenvalues into Eq. (39) would miss the derivatives and is not a valid Hessian.

8 Parallel PQ domain-wall repair

The baseline anomaly audit gave

$$\mathcal{A}_{\text{Spin}(10)^2\text{-PQ}} = -6, \quad \hat{N}_{\text{QCD}} = -12, \quad N_{\text{DW}} = \frac{12}{|\mathbb{Z}_4^{\text{diag}}|} = 3. \quad (40)$$

This agrees with the gauge-center identification emphasized in Ref. [5]. Post-inflationary PQ breaking is therefore unsafe without additional physics.

A minimal matter repair is to add two left-handed Weyl fields

$$X_1, X_2 \sim 10_F, \quad q_{\text{PQ}}(X_a) = +2, \quad \mathcal{L} \supset -\frac{1}{2} y_{ab} S X_a X_b + \text{h.c.} \quad (41)$$

The 10 is a real gauge representation, so there is no perturbative $\text{Spin}(10)^3$ anomaly. Each field contributes $qT(10) = 2$, and hence

$$\Delta\mathcal{A} = 4, \quad \mathcal{A}' = -2, \quad \hat{N}' = -4, \quad \boxed{N'_{\text{DW}} = |-4|/4 = 1.} \quad (42)$$

The diagonal quotient survives: the generator of the $\text{Spin}(10)$ center acts as -1 on 10_F , while the $\pi/2$ PQ rotation gives $e^{iq\pi/2} = e^{i\pi} = -1$.

Below M_I , a Dirac 10_F gives

$$\Delta a^{\text{SM}} = (2/3, 2/3, 2/3), \quad \Delta b^{\text{SM}} = \begin{pmatrix} 38/3 & 0 & 2/15 \\ 0 & 49/6 & 3/10 \\ 16/15 & 9/10 & 7/30 \end{pmatrix}. \quad (43)$$

Above M_I ,

$$\Delta a^{\text{PS}} = (4/3, 4/3, 4/3), \quad \Delta b^{\text{PS}} = \begin{pmatrix} 110/3 & 0 & 0 \\ 0 & 49/3 & 3 \\ 0 & 3 & 49/3 \end{pmatrix}. \quad (44)$$

Thus one-loop differential unification is unchanged, although α_U and the two-loop solution depend on $M_F \sim y_F v_s / \sqrt{2}$. We label this extension P54PQ-F10; it is not silently added to P54PQ-v2. Its mass-dependent replay is instead performed explicitly. For three benchmark values, M_F lies inside the PS interval and gives

y_F	M_F/M_U	M_I/M_U	α_U^{-1}
0.5	0.156270	0.125482	39.27980
1.0	0.313169	0.125873	39.43828
2.0	0.627590	0.126263	39.59606

The extension therefore preserves a perturbative two-stage solution across this declared mass window. It remains a separate branch because y_F is a new threshold parameter and the baseline action is frozen.

Pre-inflationary PQ breaking remains a cosmological boundary condition, not a particle-theory repair: it requires reheating, abundance, initial-angle, and isocurvature inputs. A small explicit bias likewise requires a common window satisfying both wall decay and $|\bar{\theta}| < 10^{-10}$; recent work illustrates mechanisms of this type [6], but no such operator is promoted here.

9 Bosonic Coleman–Weinberg matching addendum

The dedicated follow-up calculation uses background-field Landau gauge and $\overline{\text{MS}}$ at $\mu_U = g_U \omega$. The hard spectral functional retains 290 positive scalar modes and 33 positive vector modes. The 38 scalar and 12 vector infrared modes remain in the low-energy theory, while Landau ghosts have $m_{\text{gh}}^2 = \xi_{\text{gf}} M_V^2 = 0$ and no hard term. The Fréchet second variation of the complete field-dependent scalar and vector mass operators gives

$$\begin{aligned}\Pi_s^{\text{hard}}/\omega^2 &= -0.0182525705951 \mathbf{1}_4, \\ \Pi_v^{\text{hard}}/\omega^2 &= -0.00198408049986 \mathbf{1}_4, \\ \Pi_B^{\text{hard}}/\omega^2 &= -0.0202366510950 \mathbf{1}_4.\end{aligned}\tag{45}$$

Since the light vector has 10_H weight $w_{10} = 0.999997474290$, the bosonic retuning is

$$\delta\xi_{02,B}(\mu_U) = -0.0202367022071.\tag{46}$$

This is a scheme-dependent zero-momentum matching curvature, not a pole mass. The unfit heavy-Yukawa projection is kept as

$$\eta_Y(\mu) = \frac{\langle h | \Pi_{\text{heavy-Y}}(0; \mu) | h \rangle}{\omega^2}, \quad \delta\xi_{02}(\mu) = \frac{\kappa_B(\mu)/\omega^2 + \eta_Y(\mu)}{w_{10}}.\tag{47}$$

The full derivation, scale replay, and numerical checks are in `tex/p54_p2.bosonic_cw.tex`.

10 Gate status and next calculation

Gate	Status
two-loop SM/PS engine and finite matching	closed
historical no-threshold reproduction	closed
all 35 P1 rows and scalar threshold Jacobian	closed
historical collapsed covariance regression	closed, diagnostic only
PS parent projectors and two-site spectral traces	closed
hierarchy/threshold fixed point	closed at $r = 0.1265$
two-site experimental and log-mass covariance	closed
loop retuning theorem, condition, and heavy gap	closed
bosonic zero-momentum CW matching curvature	closed in declared scheme
heavy-Yukawa projection	explicit P3 matching nuisance
gauge-independent pole mass	not claimed
baseline $N_{\text{DW}} = 3$ audit	closed, unsafe post inflation
candidate $10_F + 10_F$ repair to $N_{\text{DW}} = 1$	derived; three two-loop mass benchmarks close
P2 bosonic matching gate	closed

The hierarchy-restored P1 search, two-site P2 replay, and bosonic CW coefficient are complete. P3 may now profile the explicit nuisance $\eta_Y(\mu)$ and must test the corrected heavy-doublet norm against the $0.0305570303\omega^2$ gap. Neither $\eta_Y = 0$ nor a gauge-independent pole mass is inferred from the vacuum spectrum.

References

- [1] M. E. Machacek and M. T. Vaughn, “Two-Loop Renormalization Group Equations in a General Quantum Field Theory,” Nucl. Phys. B **222**, 83 (1983); **236**, 221 (1984); **249**, 70 (1985).
- [2] K. S. Babu and S. Khan, “A Minimal Non-Supersymmetric SO(10) Model: Gauge Coupling Unification, Proton Decay and Fermion masses,” [Phys. Rev. D **92**, 075018 \(2015\)](#).
- [3] N. Haba, Y. Shimizu, and T. Yamada, “Neutrino Mass in Non-Supersymmetric SO(10) GUT,” [arXiv:2304.06263](#).
- [4] S. Navas et al. (Particle Data Group), “Review of Particle Physics,” [Phys. Rev. D **110**, 030001 \(2024\)](#).
- [5] A. Ernst, A. Ringwald, and C. Tamarit, “Axion Predictions in SO(10) $\times$ U(1)_{PQ} Models,” [JHEP **02** \(2018\) 103](#).
- [6] Y. Zhang, “Imperfect Axion Precludes the Domain Wall Problem,” [arXiv:2305.15495](#).